

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Mining Engineering

2. Subject Name: Planning and Design Laboratory **L-T-P:** 0-0-3 **Credits:**2

3. Pre-requisites: NIL

4. Syllabus and reference books:

Syllabus: Drill hole database preparation, Solid modeling, block modeling, conventional resources estimation-ISD, ID, geostatistics-variogram modeling, kriging, Pit Design, Pushback design and production scheduling, Haul road design, underground stope design, pit slope analysis, pillar design

Reference Books:

1. SURPAC Manual
2. Whittle Manual
3. Minex Manual
4. Minesight Manual

5. List of Experiments:

Sl. No.	Experiment Name
1.	Introduction to SURPAC/MINESIGHT and Creation of Drill hole Database, visualization of drill hole layout, drill hole traces in SURPAC
2	Preparation of DTM and Solid Model, Volume calculation and Tonnage Estimation using SURPAC/MINESIGHT
3	Block Model- Part I: preparation of block model, estimation using ID and ISD,
4	Block Model-Part II: Geostatistics-Basic statistics, variogram modeling and estimation using kriging
5	Manual pit Design, Haul Road Design in SURPAC/MINESIGHT

6	Introduction to Whittle software- Importing Block model from SURPAC, Slope definition, Mining and Processing cost estimation, Design of optimal pit in Whittle
7	Production scheduling and Pushback design in Whittle
8	Introduction to MINEX software, Drill hole data base preparation, Fault Modeling and Insitu coal resource estimation
9	Pit layout design, Strip design and Haul road design in MINEX
10	Pit Slope Analysis in Geoslope/ Galena software
11	Underground pillar design and Safety Factor Analysis